Census income classification: report

UCI Adult dataset · predicting income above $50K

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1. Problem and data

The task is to predict whether a person in the 1994 US census earns more than $50K a year. I combined *adult.data* and *adult.test* into 48,842 records, read ? as missing, and made one stratified 80/20 split with random_state=42 (39,073 train, 9,769 test). The classes are imbalanced: 37,155 people earn at most $50K and 11,687 (23.9%) earn more. Missing values appear in *workclass* (2,799), *occupation* (2,809) and *native-country* (857). In the training data, the share earning >50K is 44.5% for married-civ-spouse and 4.6% for never-married people, 9.4% when *workclass* is missing, and 61.8% among people with a capital gain (91.8% have none).

Accuracy alone is misleading here: a model that always says '<=50K' scores 0.761 accuracy but finds no high earners (recall 0, F1 0). I therefore judge models by precision, recall and F1 for the >50K class, plus ROC AUC.

2. Preprocessing plan (no leakage)

One `ColumnTransformer` inside a `Pipeline`, so every statistic is fitted on the training part of each CV fold only. **Numeric** (age, education-num, hours-per-week, capital-gain, capital-loss): median imputation, then standard scaling for logistic regression only. **Categorical** (workclass, marital-status, occupation, relationship): constant 'Unknown' imputation, because missingness is informative, then one-hot encoding. **Dropped:** *fnlwgt* (survey weight), *education* (duplicates education-num), *race* and *sex* (excluded as model inputs, see the fairness section), and *native-country* (CV F1 0.656 against 0.657 with it, half the columns, and a proxy for national origin). A log transform of the capital columns lowered logistic-regression F1 (0.645 against 0.657), so they stay raw.

3. Model comparison (5-fold stratified CV on the training set)

Table 1. Default settings (mean ± std over folds). Metrics for the >50K class.

Model	Accuracy	Precision	Recall	F1	ROC AUC
Majority-class baseline	0.761 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.500 ± 0.000
Logistic Regression	0.851 ± 0.002	0.733 ± 0.005	0.594 ± 0.009	0.656 ± 0.006	0.905 ± 0.001
Random Forest	0.847 ± 0.003	0.706 ± 0.010	0.617 ± 0.005	0.659 ± 0.004	0.892 ± 0.002
Gradient Boosting	0.866 ± 0.003	0.787 ± 0.011	0.606 ± 0.011	0.685 ± 0.008	0.920 ± 0.002

Table 2. Each model tuned with its own small grid (same folds, scored on F1).

Model	Best parameters (small grid)	Precision	Recall	F1	ROC AUC
Logistic Regression	{C: 10}	0.733 ± 0.005	0.594 ± 0.009	0.656 ± 0.006	0.905 ± 0.001
Random Forest	{max_depth: 20, min_samples_leaf: 2}	0.778 ± 0.007	0.601 ± 0.013	0.678 ± 0.008	0.918 ± 0.002
Gradient Boosting	{max_depth: 5, n_estimators: 150}	0.779 ± 0.008	0.653 ± 0.006	0.711 ± 0.005	0.928 ± 0.001

Gradient boosting wins on F1 (0.711) and ROC AUC (0.928) against random forest (0.678) and logistic regression (0.656). The best settings sit on the grid edges; two neighbouring settings for gradient boosting scored 0.7109 and 0.7068 against 0.7107, so the search is flat there.

4. Feature selection

Table 3. L1 logistic regression against the L2 baseline.

Setup	C	CV F1	Non-zero coef.
L2 (baseline)	1	0.6558	42
L1	1	0.6562	35
L1	0.1	0.6536	30
L1	0.03	0.6485	27
L1	0.01	0.6343	15

L1 with C=0.1 keeps 30 of 42 coefficients for a loss of 0.002 F1, well inside the fold standard deviation (about 0.007). Putting the same L1 selection in front of gradient boosting gives F1 0.7102 against 0.7107 with all 42 columns, using about 31 inputs. I retain the selection step: the model is simpler and nothing measurable is lost.

5. Final model and test results

Final model: gradient boosting (max_depth=5, 150 trees) behind the L1 selection step. The decision threshold, 0.35, was chosen on out-of-fold predictions of the training set (F1 0.710 at 0.50, 0.727 at 0.35). The test set was evaluated once.

Precision	Recall	F1	ROC AUC	Avg. precision	Accuracy
0.692	0.766	0.728	0.930	0.831	0.863

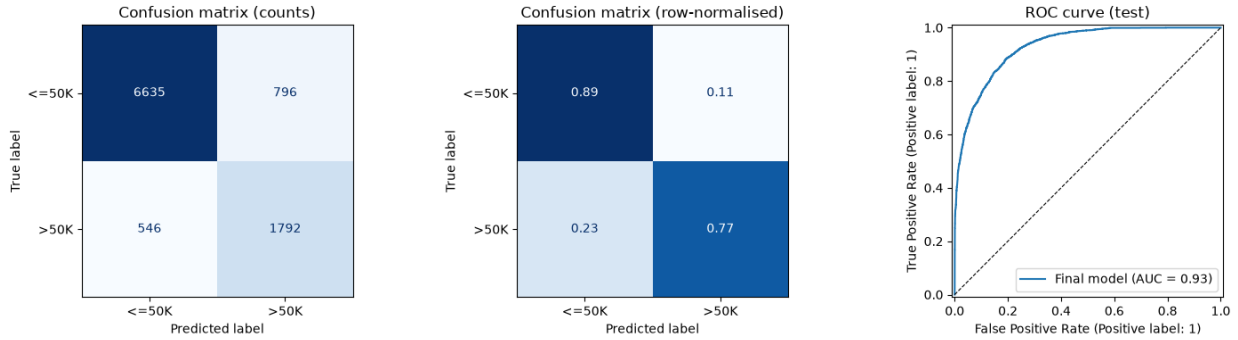


Figure 1. Test confusion matrix (TN 6,635, FP 796, FN 546, TP 1,792) and ROC curve.

Error analysis. The 796 false positives look like the true positives: 93% are married-civ-spouse, about 45 years old, working 46 hours a week, often in executive or professional jobs, with a mean predicted probability of 0.54. The 546 false negatives almost never have capital gains (29% of the true positives do), have less education (10.0 against 12.0 years) and are often craft or machine-operator workers, with a mean probability of 0.19. The model leans on marital status, capital gain and education (permutation importance 0.067, 0.051 and 0.033), so it over-flags married professionals and misses well-paid people in trades without capital income. Cross-validation (OOF F1 0.727) and test (0.728) agree.

6. Reflection

This model predicts whether a person earned more than $50K a year in the 1994 census. Its limits start with the data. The labels are over thirty years old, the threshold means little today, and "high income" is partly a household quantity here, because marital status and relationship are among the strongest signals. On the test set the final model reaches an F1 of 0.73 for the >50K class, but it still misses about one in four high earners (recall 0.77), and about three in ten of the people it flags are wrong (precision 0.69). The errors are not random: the false negatives are mostly people with no capital income and an ordinary profile, so the model has almost nothing to go on for them.

Bias is the main concern. I did not use race or sex as features, but the inspection tables show that this does not make the model neutral. Marital status and relationship work as proxies for sex, and marital status was the most important feature in the permutation test. Men are flagged as high earners far more often than women (34.3% versus 11.2%), their false positive rate is more than three times higher (14.8% versus 4.4%), and recall is higher for men (0.79 versus 0.65). Part of this mirrors real gaps in the 1994 labels, so the model learns and repeats historical inequality instead of anything about an individual. Two of the racial groups are very small (69 and 104 people), so those numbers are too noisy to conclude anything from.

I would not use this model to decide anything about real people, for example credit, hiring or benefits. Before any use, the first thing I would do is a proper fairness audit on current, representative data: compare error rates across sex and race with confidence intervals, test whether removing marital status and relationship narrows the gaps and what that costs in F1, and agree with the people who own the decision which kind of error is worse. The decision threshold (0.35 here) would have to be revisited with that cost in mind, since it deliberately trades precision for recall. If the audit fails, the model should only be used for aggregate analysis, with a human making any individual decision.

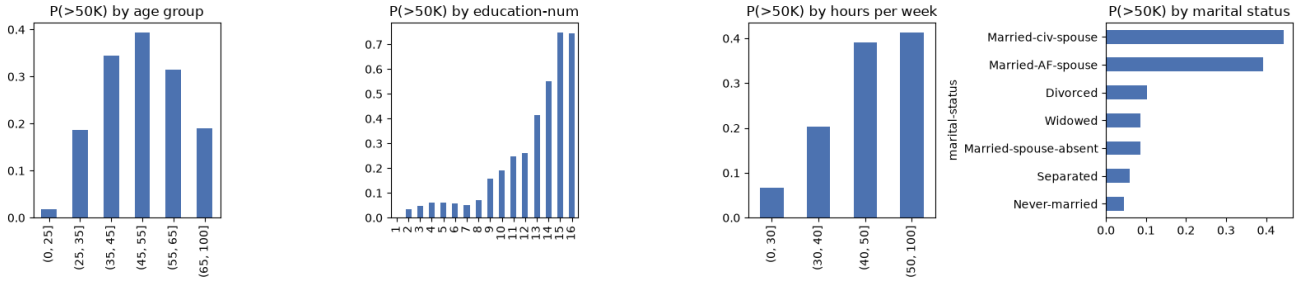
Appendix

A. Fairness inspection on the test set (sex and race were not features)

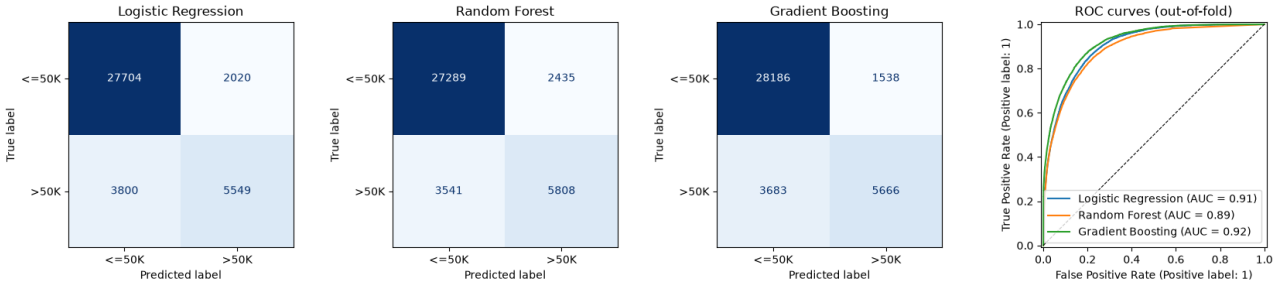
Group	n	Actual >50K	Predicted >50K	Precision	Recall	FPR
Male	6480	0.304	0.343	0.699	0.788	0.148
Female	3289	0.112	0.112	0.651	0.651	0.044
White	8348	0.254	0.280	0.701	0.774	0.112
Black	944	0.128	0.137	0.612	0.653	0.061
Asian-Pac-Islander	304	0.257	0.296	0.656	0.756	0.137
Amer-Indian-Eskimo	104	0.087	0.173	0.333	0.667	0.126
Other	69	0.159	0.159	0.636	0.636	0.069

Groups with fewer than about 150 people (Amer-Indian-Eskimo, Other) give very noisy rates.

B. Exploration of the target (training data)



C. Cross-validated confusion matrices and ROC curves (out-of-fold, training set)



D. Reproducibility

Python 3.14.7, scikit-learn 1.9.1, pandas 3.1.0, numpy 2.5.4 (full list in requirements.txt). random_state=42 everywhere; the notebook runs top to bottom and writes the numbers in this report to results.json.

Data source. Becker, B. and Kohavi, R. (1996). Adult. UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/2/adult